

OPERATING MANUAL

SPINDLE APPLICABLE TO BIG-PLUS HOLDER

Precautions on the use of the
BIG-PLUS holder

Manual No.: SGMCSB0072E

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. This manual was considered complete and accurate at the time of publication, however, due to our desire to constantly improve the quality and specification of all our products, it is subject to change or modification. If you have any questions, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.

Issued by Manual Publication Section, YAMAZAKI MAZAK CORPORATION, Japan

09.2020

Serial No. 368959
Copyright (c) 2013 YAMAZAKI MAZAK CORPORATION. All rights reserved.

CONTENTS

1	SAFETY PRECAUTIONS.....	1-1
1-1	Rules	1-1
1-2	Safety Precautions Relating to the Function/Device Described in This Manual.....	1-1
2	PRECAUTIONS ON THE USE OF THE BIG-PLUS HOLDER.....	2-1

Serial No. 368959
Copyright (c) 2013 YAMAZAKI MAZAK CORPORATION. All rights reserved.



Serial No. 368959
Copyright (c) 2013 YAMAZAKI MAZAK CORPORATION. All rights reserved.

1 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

1-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:



DANGER

: Failure to follow these instructions could result in loss of life.



WARNING

: Failure to observe these instructions could result in serious harm to a human life or body.



CAUTION

: Failure to observe these instructions could result in minor injuries or serious machine damage.

1-2 Safety Precautions Relating to the Function/Device Described in This Manual



CAUTION

- Always keep clean the inside of the coolant cover, especially, the spindle taper and edge and the magazine interior (such as the tool pockets and shifter).
If chips are left in these sections, the likely ingress of dirt from the clearance occurring when a tool is not mounted or when a tool holder other than the BIG-PLUS holder is used will damage the taper and edge of the spindle, resulting in abnormal operation of the machine.
- As for machines the spindle of which has an air blow hole at the edge, if a tool holder other than the BIG-PLUS holder is used, coolant may get into the spindle through the air blow hole during machining. To avoid this, be sure to use the BIG-PLUS holder.

1

SAFETY PRECAUTIONS



Serial No. 368959
Copyright (c) 2013 YAMAZAKI MAZAK CORPORATION. All rights reserved.

2 PRECAUTIONS ON THE USE OF THE BIG-PLUS HOLDER

This machine has a spindle applicable to the BIG-PLUS spindle system. This spindle has dihedral restraint structure, in which the taper and edge of the spindle come into contact at the same time.

Before using the BIG-PLUS holder, be sure to remove dirt from the tapers and edges of the spindle and tool using a clean cloth to maintain the dihedral restraint of the BIG-PLUS spindle system.



- Always keep clean the inside of the coolant cover, especially, the spindle taper and edge and the magazine interior (such as the tool pockets and shifter).
If chips are left in these sections, the likely ingress of dirt from the clearance occurring when a tool is not mounted or when a tool holder other than the BIG-PLUS holder is used will damage the taper and edge of the spindle, resulting in abnormal operation of the machine.
- As for machines the spindle of which has an air blow hole at the edge, if a tool holder other than the BIG-PLUS holder is used, coolant may get into the spindle through the air blow hole and from the clearance shown in Fig. 2-1 during machining. To avoid this, be sure to use the BIG-PLUS holder.

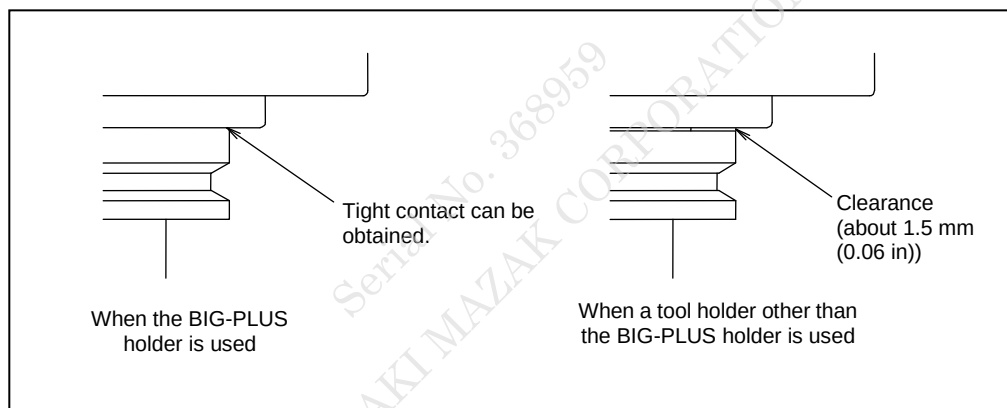


Fig. 2-1 Clearance occurring when tool holders are mounted



Serial No. 368959
Copyright (c) 2013 YAMAZAKI MAZAK CORPORATION. All rights reserved.